MANIPAL INSTITUTE OF TECHNOLOGY
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UrjaGrid Monitor

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INTRODUCTION

- Small-scale solar systems often lack intelligent control over limited battery capacity.
  This makes it difficult to monitor energy generation, battery status, and load consumption in real time.
- Efficient energy management is important to prevent unnecessary battery discharge.
- Our project develops a **20 W solar microgrid with a 12 V SLA battery** for smart energy management.
- Using **ESP32, sensors, relays, and Blynk**, the system monitors conditions and prioritizes essential loads.
- Non-critical loads are automatically disconnected when the battery level is low or unsafe conditions occur.



WORKING PRINCIPLE

20 W Solar Panel → Generates renewable energy.

Charge Controller + 12 V SLA Battery → Regulates and stores energy.

INA219 + DS18B20 → Measures voltage, current, power and temperature.

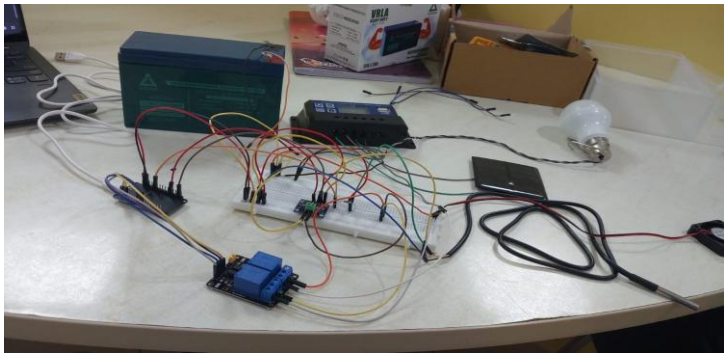
ESP32 → Processes data and controls the system.

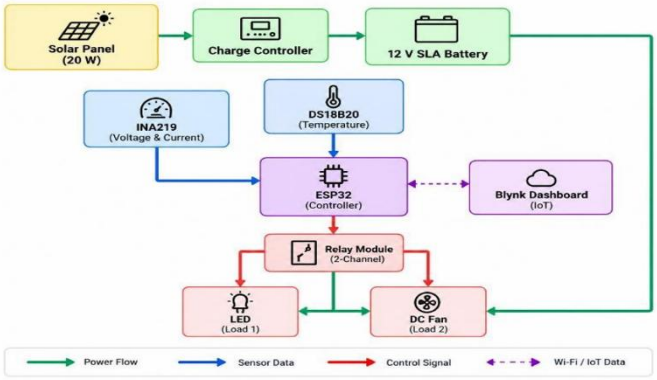
Relay + LED/Fan → Manages loads based on battery and safety conditions.

Blynk Dashboard → Displays system status, graphs and alerts.

Flow:

Solar Panel → Charge Controller → Battery → Sensors → ESP32 → Relay → Loads → Blynk Dashboard





APPLICATIONS

- **Off-grid solar systems** → Powers basic loads in remote areas.
- **Smart home energy management** → Monitors and controls energy consumption.
- **Rural electrification** → Supports lighting and essential electrical loads.
- **Renewable energy research & education** → Demonstrates IoTbased energy monitoring and intelligent load control.

STANDARDS

- **IEC 60364-7-712:2025** → Requirements for solar PV electrical installations and energy storage.
- **IEC 62477-1:2022** → Safety requirements for power electronic systems, including protection and monitoring.
- **ISO 50001:2018** → Framework for energy management and improving energy efficiency.

ADVANTAGES

- **Real-time monitoring** of voltage, current, power and temperature.
- **Intelligent load prioritization** saves battery energy.
- **Automatic load shedding and reconnection** based on battery conditions.
- **IoT-based remote monitoring** through the Blynk dashboard.
- **Low-cost educational prototype** for renewable energy and IoT applications.

FUTURE PROSPECTS

- **Scale up** for larger solar and battery systems.
- **Add more sensors and loads** for advanced energy management.
- **Integrate DSA** for energy prediction and optimization.
- **Support remote/off-grid applications** such as rural lighting.
- **Expand to smart homes and microgrid networks.**

CONCLUSION

This project demonstrates a **smart IoT-based solar microgrid** with real-time monitoring and intelligent load control.
It improves **energy utilization, battery protection, and load management**.
The prototype provides a practical foundation for **future scalable and smarter renewable energy systems**.